

Types of predictability

Video transcript

Predicting the weather and climate is a longstanding challenge in physical science and is affected by a complex range of climate and weather systems. These challenges can be separated into two main groups. The initial condition problems and the boundary condition problems.

Initial condition problems rely on knowledge about the existing state of the climate and how it evolves overtime. A classic example being a day ahead weather forecast. Whereas, boundary condition problems rely on the changes in the forcing affecting the climate. Such as climate change projections many decades into the future. To illustrate this, we can think of the climate system as a pegboard. Each ball represents a weather day and the bucket it falls into represents the weather outcome. With the initial condition known, we can accurately predict which bucket the ball will fall into. If the ball is dropped slightly left of centre, it tends to fall left or cold. If dropped right of centre, it tends to fall right or warm. The more we know about the initial conditions, the more likely we are to accurately predict its outcome.

For longer range forecasts, the predictive advantage of knowing the initial condition diminishes. We can see this here with the addition of more rows. In the context of the atmosphere, predictability of this type is typically limited to within a few days of the forecasts launch. When predicting boundary condition climate change, the concern is no longer predicting an individual weather event or in this case, the path of an individual ball. But rather the climate distribution produced by many weather events represented by the pattern made by many balls once they've dropped to the bottom. Here predictability arises due to a change in the forcing. For example, an increase in greenhouse gases indicates that surface temperatures will rise. This can be illustrated by tipping the pegboard. The balls are now more inclined to fall to the right, skewing the outcome.

More recently, climate science has seen considerable developments in the middle ground between boundary and initial climate predictions. This middle ground is often referred to as sub-seasonal, seasonal or decadal forecasting. In this range, weeks to years ahead, initial condition predictability of the fast evolving atmosphere is lost. But boundary condition predictability can be derived from the initial conditions of more slowly evolving components of the climate system. Such as the oceans, land surface or upper atmosphere.

In conclusion, there are many opportunities for exploiting weather and climate predictability across many different timescales. These predictions can take different forms and may rely on different components of the climate system depending on the time scale considered. Nevertheless, predicting weather and climate remains a major challenge for physical science and is likely to remain so for the foreseeable future.